Grade 11, Term 2, Lesson 1, Criterion C1

# Parameters and sample statistics

**C1.** Distinguishes population parameters from sample statistics by interpreting and comparing the notation for population and sample means, standard deviations, and proportions in economic and financial contexts.

# Highlight legend

| Sample (green) | Population (red) |
|---|---|
| Sample mean ($\bar{x}$) | Population mean ($\mu$) |
| Sample standard deviation ($s$) | Population standard deviation ($\sigma$) |
| Sample proportion ($\hat{p}$) | Population proportion ($p$) |
| Sample (selected units) | Population (all units of interest) |

# Which electricity bills are we studying?

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**Predict.** Which bills might the utility want to describe? Which bills could it use to estimate the mean charge?

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**Predict.** Which bills might the utility want to describe? Which bills could it use to estimate the mean charge?

**Target population.** All May electricity bills for small shops in one district.

**Sample.** 40 randomly selected bills from that list.

# From electricity bills to a mean

**Predict.** Which group does each mean describe?

| Population parameter | Sample statistic |
| --- | --- |
| | |

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| $\mu$ is the unknown mean charge across all May bills for small shops in the district. | $\bar{x} = 210$ thousand COP is the mean charge across the 40 selected bills. |

The sample mean $\bar{x}$ estimates the population mean $\mu$.

# Quick check. Redraw or redefine?

**Predict.** Which population mean does each draw estimate?

| New draw |
| --- |
| 40 bills from the original May list |
| 40 bills from the combined May list for small shops in two districts |

# Quick check. Redraw or redefine?

**Predict.** Which population mean does each draw estimate?

| New draw | Suggested answer |
|---|---|
| 40 bills from the original May list | A fresh draw can give a different $\bar{x}_{may}$ while estimating the same $\mu_{may}$ for the original district. |
| 40 bills from the combined May list for small shops in two districts | |

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| 40 bills from the combined May list for small shops in two districts | This draw's mean $\bar{x}_{2D}$ estimates $\mu_{2D}$, the population mean across both districts. |

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| 40 bills from the combined May list for small shops in two districts | This draw's mean $\bar{x}_{2D}$ estimates $\mu_{2D}$, the population mean across both districts. |

The two population means could have the same numerical value.

# A market association's sales survey (C1)

A market association wants to describe all 1,200 stalls licensed for June. It randomly selects 60 stalls.

In the survey, mean June sales revenue is 4.8 million COP and the standard deviation is 1.2 million COP. 18 sampled stalls accept digital payment.

**Task 1.** Which stalls make up the population? Which make up the sample?

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**Task 1.** Which stalls make up the population? Which make up the sample?

**Population.** All 1,200 June-licensed stalls.

**Sample.** The 60 randomly selected stalls.

# Match each quantity to its symbol

**Task 1, continued.** Which sample statistic estimates each population parameter? What does each symbol describe?

| Population parameter | Sample statistic |
| --- | --- |

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| Population parameter | Sample statistic |
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| $\mu$, mean June sales across all licensed stalls | $\bar{x}_1 = 4.8$ million COP, mean June sales across the 60 |

# Match each quantity to its symbol

**Task 1, continued.** Which sample statistic estimates each population parameter? What does each symbol describe?

| Population parameter | Sample statistic |
|---|---|
| $\mu$, mean June sales across all licensed stalls | $\bar{x}_1 = 4.8$ million COP, mean June sales across the 60 |
| $\sigma$, standard deviation of June sales across all licensed stalls | $s = 1.2$ million COP, standard deviation of June sales across the 60 |

# Match each quantity to its symbol

**Task 1, continued.** Which sample statistic estimates each population parameter? What does each symbol describe?

| Population parameter | Sample statistic |
|---|---|
| $\mu$, mean June sales across all licensed stalls | $\bar{x}_1 = 4.8$ million COP, mean June sales across the 60 |
| $\sigma$, standard deviation of June sales across all licensed stalls | $s = 1.2$ million COP, standard deviation of June sales across the 60 |
| $p$, fraction of all June-licensed stalls accepting digital payment | $\hat{p} = 0.30$, fraction accepting digital payment among the 60 |

# The sample share accepting digital payment

**Predict.** What does each count in the fraction describe?

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**Numerator.** 18 sampled stalls accept digital payment.

**Denominator.** All 60 sampled stalls are counted.

The sample proportion is unitless.

# A fresh draw from the June list

**Population.** All 1,200 June-licensed stalls. The association takes a fresh random sample from the same list to see how survey results vary across draws.

- Mean June sales revenue is 5.1 million COP.
- Standard deviation is 1.0 million COP.
- Digital-payment share is 0.25 = 25%.

**Task 2.** Revisit your answers to Task 1. What changes, what stays fixed, and which differences can you calculate?

# Compare the two June draws

**Population.** All 1,200 June-licensed stalls.

**Predict.** Which sample statistics changed across the two draws?

| Sample statistic | First draw | Second draw |
|---|---|---|
| June sales mean (million COP) | | |
| June sales standard deviation (million COP) | | |
| Digital-payment share (unitless) | | |

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**Predict.** Which sample statistics changed across the two draws?

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| June sales mean (million COP) | $\bar{x}_1 = 4.8$ | $\bar{x}_2 = 5.1$ |
| June sales standard deviation (million COP) | $s = 1.2$ | $s = 1.0$ |
| Digital-payment share (unitless) | $\hat{p} = 0.30$ | $\hat{p} = 0.25$ |

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| Digital-payment share (unitless) | $\hat{p} = 0.30$ | $\hat{p} = 0.25$ |

All three statistics differ in this fresh draw.

# Second draw minus first draw

**Predict.** What are the three differences between sample statistics?

| Sample statistic | Difference between draws |
|---|---|
| Mean | |
| Standard deviation | |
| Digital-payment share | |

**Predict.** What stays fixed?

# Second draw minus first draw

**Predict.** What are the three differences between sample statistics?

| Sample statistic | Difference between draws |
|---|---|
| Mean | $\bar{x}_2 - \bar{x}_1 = 5.1 - 4.8 = +0.3$ million COP |
| Standard deviation | |
| Digital-payment share | |

**Predict.** What stays fixed?

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**Predict.** What stays fixed?

The population parameters $\mu$, $\sigma$ and $p$ stay fixed for the same June population.